

3D AI People Counter

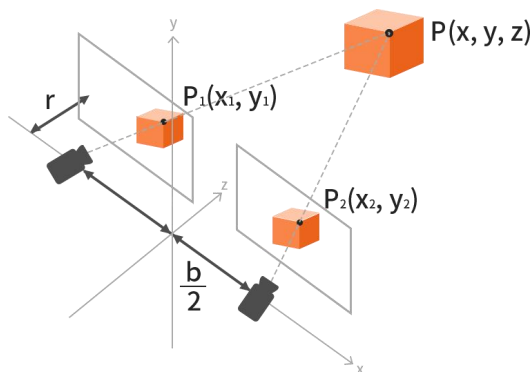
HX-CCD21 (AI Version)

Effective customer flow · Edge AI · ReID deduplication



Core Technology

The HX-CCD21's ability to output "effective foot traffic" stems from the synergy of four core technologies. This is not a simple counter upgrade, but an entirely new data category.



3D Binocular Imaging 99% Counting Accuracy

Binocular 3D stereoscopic vision precisely locates human 3D position, **with an accuracy rate of up to 99%**; accurately filters interference from children, luggage, shopping carts, etc., resists light interference, and operates stably 24/7.

Edge AI · 2 TOPS Computing Power Engine

Built-in **2TOPS heterogeneous computing power**, millisecond-level local AI inference, **no external edge box required**; video processed locally without uploading to the cloud, balancing real-time performance with privacy and security.

ReID Full-Body Feature Deduplication · Eliminates Duplicate Counts

Utilizes **faceless** human feature ReID deep learning for automatic deduplication. Repeated entry and exit by riders or staff **are not counted multiple times**; accurately records customer arrival and departure times to calculate dwell time.

Intelligent Recognition of Staff/Delivery/Riders

Badge recognition filters out staff (80% accuracy), combined with ReID feature matching to distinguish riders and customers, **fully automatic recognition** with no manual operation required.

Why is "Effective Foot Traffic" a New Category?

Traditional devices miscount staff, delivery riders, etc., inflating foot traffic and misleading judgments on conversion rates, staffing, and marketing ROI. The HX-CCD21 edge AI automatically filters non-customer noise, outputting true effective foot traffic—enabling stores to see their real transaction capabilities for the first time.

Feature Comparison



Comparison Dimension	Traditional Statistics	HX-CCD21
Counting Method	Counts heads; all entries and exits are counted.	3D Binocular + AI, precise recognition
Staff Filtering	Cannot recognize; all are counted.	Badge recognition automatically excludes.
Delivery/Courier Filtering	Repeatedly counts as they move back and forth.	ReID deduplication, no repetition
Multi-Entrance Stores	Independent counting at each entrance.	Multiple units linked for unified deduplication.
Dwell Time	/	Precise recording of arrival and departure times.
Accuracy Rate	Inflated; noise rate up to 58.8%.	99% counting accuracy.
Demographics	/	Gender 95% / Age 80%
Privacy Compliance	Some require facial capture.	Zero facial capture, local computation.

Product Value

The HX-CCD21 is more than just a counter; it transforms the data foundation for store operational decisions. When foot traffic data becomes accurate, six core operational capabilities are upgraded.



True Conversion Rate No longer diluted.

The denominator returns to the actual customer count, so conversion rates are no longer diluted by noise from staff, delivery personnel, or couriers.

See the store's true transaction capability for the first time.

—Diagnostic focus shifts from "poor transactions" to "insufficient foot traffic" or "conversion rate needs improvement."



Precise Staff Scheduling Labor costs reduced

Eliminates false peaks created by delivery riders, restoring the true customer arrival curve.

Staff appropriately during the right times—avoiding wasted labor costs and ensuring sufficient staff during real peak hours to prevent lost sales. Tested to reduce labor costs by 15% in chain pharmacies.



Marketing True ROI Every dollar counts.

Use the actual customer count as the baseline for comparison before and after promotions. Marketing incremental gains are clearly visible.

Accurately calculate the true return on every marketing investment for the first time, ensuring marketing budgets are spent effectively.



Dwell Time Analysis Customer Segmentation Insights

Based on ReID data, records the arrival and departure times of each effective customer to precisely calculate in-store dwell time.

Automatically categorizes customers by dwell time: passersby (<3 minutes), browsers (3-10 minutes), potential buyers (>10 minutes), to optimize displays and sales pitches accordingly.



Customer Profiling Precise Marketing Decisions

Gender recognition accuracy is 95%, age recognition accuracy is 80%, generating rich customer profiles.

Know your customers' gender and age group—product selection, display, and promotion strategies are data-supported, no longer based on guesswork.



Multi-Store Comparison Benchmark for Chain Management

With a unified data baseline, chain stores become truly comparable. Which stores have high conversion rates, staffing issues, or effective marketing—is clear at a glance.

Use data to find benchmarks, use benchmarks to guide the entire operation. Chain management evolves from experience-driven to data-driven.

Product Functions



Privacy Compliance

It does not extract facial information and performs all calculations locally, complying with the requirements of the Personal Information Protection Law.



Data Security

Local flash storage for 90 days, resume download after network outage, encrypted data transmission, zero risk of data loss.



Data Upload Timeliness

Upload time can be customized, supporting real-time, 1 minute, 5 minutes, 30 minutes, etc.



Enrich Open Interfaces

The device offers a rich set of locally available application interfaces, supports RS-485 extensions, and allows for flexible and rapid integration.



All-environmental Adaptability

IP66 dustproof and waterproof, wide operating temperature range of -10°C to 50°C, stable operation in both bright outdoor light and dim lighting environments.



Free Cloud Platform

It comes with a smart retail cloud management platform, providing real-time dashboards, multi-dimensional reports, and multi-store management, all free of charge.



Multi-protocol Support

HTTP POST / HTTPS POST / FTP / SFTP / MQTT, supporting secondary development.



Professional After-sales Service

A professional technical team provides 24/7 support, including remote diagnostics and on-site service, ensuring stable equipment operation.

Functional Mode



Wheelchair Mode

It can identify wheelchair users individually. Quantity, and give a signal.



Kids Mode

Once a child's height is set, anyone shorter than that height will be considered a child.



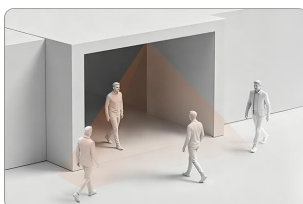
Control Mode

The equipment is paired with relays, Control multiple electrical appliances.



Data Display Mode

The equipment is paired with the occupancy early warning host. Visualizing data.



Channel Mode

It can detect the number of people passing through the entrance. and number of people entering the store.



Single-player Mmode

The device can store local data for 90 days, supports export, and can be used even without a network connection.

Data Display



Traditional customer flow statistics "count people"; **FOORIR** truly identifies "customers".

People Counting Platform

The device uploads data to the people counting platform for visualized management. The cloud platform can view data every 5 minutes, every half hour, every hour, every day, every week, every month, and every year, and generate multi-dimensional analysis reports that can be exported to Excel files to help make accurate operational decisions.



Intelligent customer group analysis

- Gender/Age
- Adults/Children
- Employees/Delivery
- Repeat visitors



Real-time passenger flow monitoring

- Real-time entering and exiting
- Average length of stay
- Thermodynamic matrix diagram
- Exporting report data



Multidimensional data analysis

- Peak and low peak analysis
- Comparative analysis
- Holiday Analysis
- Ranking Analysis

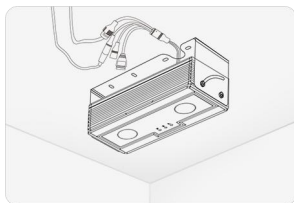


Unified management of multiple devices

- Multi-level organizational structure
- Multi-entry linkage
- Unified management of multiple stores
- Supports data integration

Minimal Installation

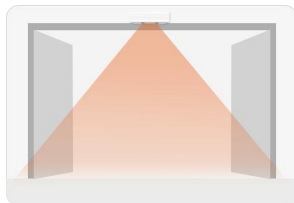
From installation to seeing the data, it only takes 10 minutes. No professional IT personnel are needed; the store's electrician can complete the installation.



1

Fixed equipment

Ceiling mount/ceiling mount/embedded mount, PoE power supply with just one network cable.



2

Software zoom

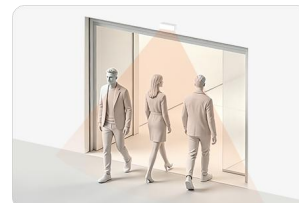
The app adjusts the coverage area to adapt to different door widths.



3

Connect to platform

Scan the QR code to bind your account and view your data instantly on the free cloud platform.



4

Start statistics

Automatic calibration, no debugging required, instantly outputs valid passenger flow.

Installation is zero-barrier.

PoE power supply (one network cable provides both power and internet access) or DC 9~36V power supply; software-based automatic zoom; adaptive installation height from 1.9 to 6 meters; resume upload after network outage; 90-day local caching; automatic upload upon network recovery. Installation can be completed without shutting down the store.

Application Scenarios



Wholesale Snacks



Freshly Made Tea Drinks



Coffee Chain



Chain Pharmacies



Beauty and Personal Care



Western Fast Food



Athletic shoes and Apparel



Casual Wear



Accessible Luxury Clothing



Convenience Store



Braised Food



Eyewear Chain



Mother and Baby Products



Supermarkets



Jewelry and Accessories



Shopping Center

Client Cases

The following three industry case studies are all based on real-world store deployments. Each case study demonstrates a comparison of effective customer traffic analysis using traditional statistics versus the HX-CCD21.

Daily data correction for bulk snack chain stores

Chain snacks - Multi-store master-slave architecture

Traditional Statistics

347

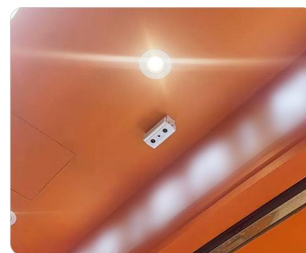
HX-CCD21 Effective Foot Traffic

143

Noise Rate

58.8%

"Before deployment, we always thought the customer traffic was okay, but the conversion rate was low. After installing the HX-CCD21, we found that the effective customer traffic was 143 people - nearly 60% of the 'customer traffic' was not customers at all. We had been making business decisions based on incorrect data."



500 chain pharmacies - 3 consecutive days of verification

Chain pharmacies - 3 consecutive days of testing

Traditional Statistics

2025

HX-CCD21 Effective Foot Traffic

921

Noise Rate

54.5%

"Pharmacies have employees, delivery personnel, and repeat customers coming and going every day, and traditional equipment counts them all. After three consecutive days of verification, the HX-CCD21 identified only 921 valid customers—more than half of the 'customers' were not actual customers buying medicine. Now, we can finally evaluate store performance based on real customer flow data and optimize scheduling, marketing, and product allocation."



Client Cases

The Truth Behind 7 Days of Data on a Century Old European Apparel Brand

Affordable luxury apparel - 7-day continuous testing

7-day Traditional Statistical Total

1028

HX-CCD21 Effective Foot Traffic

426

Noise Rate

57.5 %

"The store manager kept reporting that 'customer traffic seemed to be decreasing,' but traditional equipment data showed normal figures. After 7 days of continuous testing, the HX-CCD21 recorded 426 effective customers—more than half of the count was noise. Now we are using effective customer traffic data to reassess the true performance of each store."



The common truth in the three cases

Whether it's snacks (58.8%), clothing (57.5%), or pharmacies (cross-point matching 98.6%), the data all tell the same thing: the effective customer flow data of HX-CCD21 is consistent, while the noise rate of traditional customer flow statistics generally exceeds 50%. You don't lack customer flow data; you lack real customer flow data.

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